

CSE 566 Midterm

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I. QUESTION (14 POINTS)

In 2-fold cross-validation, the dataset X is divided randomly into 2 equal sized parts, X_1 and X_2 . For the first fold, the train and test sets are then X_1 and X_2 respectively. For the second fold, the train and test sets are then X_2 and X_1 respectively. On the other hand, classes must be represented in the right proportions not to disturb the class prior probabilities (**stratification**). Propose an algorithm to do 2-fold cross-validation on a 2-class dataset with stratification.

II. QUESTION (12 POINTS)

Let x have a Rayleigh density, for all $x \geq 0$

$$p(x|\theta) = 2\theta x e^{-\theta x^2}, \quad (1)$$

Suppose that n samples $x_1, x_2, \dots, x_n$ are drawn independently according to $p(x|\theta)$. Show that the maximum likelihood estimate for θ is given by

$$\hat{\theta} = \frac{n}{\sum_{k=1}^n x_k^2} \quad (2)$$

III. QUESTION (12 POINTS)

Suppose that the task is to cluster the following eight points (with (x, y) representing location) into three clusters. $A_1(2, 10)$, $A_2(2, 5)$, $A_3(8, 4)$, $A_4(5, 8)$, $A_5(7, 5)$, $A_6(6, 4)$, $A_7(1, 2)$, $A_8(4, 9)$. The distance function is Euclidian distance. Suppose initially we assign A_2 , A_5 , and A_8 as the center of each cluster, respectively. Run the k-means algorithm one epoch on this data.

IV. QUESTION (14 POINTS)

Let say we have one variable x , three input points (x_1, y_1) , (x_2, y_2) , (x_3, y_3) we want to make a linear fit on them, namely

$$f(x) = w_0 + w_1 x$$

Given these 3 input points, calculate the formula for the optimum w_0 and w_1 .

V. QUESTION (14 POINTS)

Give an example feature selection run on 4 feature dataset, where the best feature subset found by Forward Selection and Backward Elimination is $\{F_2, F_3\}$. On the other hand, arrange the error rates of the Feature subsets so that the global optimum is $\{F_1, F_3\}$.

VI. QUESTION (10 POINTS)

Let x have an exponential density, for all $x \geq 0$

$$p(x|\theta) = \theta e^{-\theta x}, \quad (3)$$

Show that the exponential density is normalized.

VII. QUESTION (12 POINTS)

Assume that you have a 2 dimensional 2 class dataset. For the following algorithms, give an example estimated covariance matrix S (or matrices S_i depending on the algorithm). Also plot the covariance matrices.

- Quadratic Discriminant Analysis
- Linear Discriminant Analysis
- Naive Bayes
- K-means

VIII. QUESTION (12 POINTS)

Assume that you have the following dataset.

x_1	x_2
2	4
6	5
4	3
4	8

Estimate the mean vector $\mathbf{m}$ and covariance matrix $\mathbf{S}$ for this dataset.